

Marketing Funnel & Lead Conversion Analysis Report

Project Objective

The objective of this project is to analyze customer behavior across the marketing funnel to understand how users move from visitors to leads and finally to customers. This analysis helps businesses identify drop-off points, improve conversion rates, optimize marketing campaigns, and make data-driven business decisions.

Dataset Information

The dataset contains customer interaction and transaction records collected from an e-commerce platform.

Total Records: 10,000

Total Columns: 12

Important Columns:

- event_time – Timestamp of user activity
- event_type – User action such as view, cart, or purchase
- product_id – Unique product identifier
- category_code – Product category
- brand – Product brand name
- price – Product price
- user_id – Unique customer identifier
- user_session – User session ID

Project Methodology

1. Data cleaning and preprocessing were performed using Python and Pandas.
2. Missing values and duplicate records were handled.
3. Funnel stages were mapped as:
 - Visitor → View Event
 - Lead → Cart Event
 - Customer → Purchase Event
4. Conversion rates and drop-off percentages were calculated.
5. Product category and brand performance were analyzed to identify business opportunities.

Key Funnel Metrics

Metric	Value
Total Visitors	9,682
Total Leads (Cart Events)	140
Total Customers (Purchase Events)	178

Visitor → Lead Conversion Rate	1.45%
Lead → Customer Conversion Rate	127.14%
Overall Funnel Conversion Rate	1.84%

Key Insights

- The platform recorded **9,682** visitor interactions, indicating strong overall traffic and customer engagement.
- A total of **140** users added products to the cart, producing a visitor-to-lead conversion rate of **1.45%**.
- The lead-to-customer conversion rate was calculated at **127.14%**, highlighting customer purchase completion performance.
- The overall funnel conversion rate is **1.84%**, showing the percentage of total visitors converted into customers.
- Funnel drop-offs suggest that some users lose purchase intent during the cart or checkout stages.
- High-performing categories and brands contribute significantly to total purchase activity and revenue generation.

Top Purchased Categories

- electronics.smartphone: 83 purchase events
- electronics.audio.headphone: 8 purchase events
- appliances.environment.vacuum: 6 purchase events
- electronics.video.tv: 5 purchase events
- appliances.kitchen.washer: 5 purchase events

Top Revenue-Generating Brands

- apple: Revenue generated = 25969.03
- samsung: Revenue generated = 10411.52
- huawei: Revenue generated = 3562.51
- xiaomi: Revenue generated = 1730.37
- lucente: Revenue generated = 1604.93

Business Recommendations

- Improve product page quality, descriptions, and user interface to encourage more users to add products to the cart.
- Simplify the checkout process to reduce cart abandonment and increase purchase completion rates.
- Use remarketing campaigns and cart reminder notifications to recover lost customers.
- Focus marketing efforts on high-performing categories and brands to maximize revenue growth.
- Analyze user activity trends and optimize marketing campaigns during peak engagement periods.
- Provide personalized product recommendations to improve customer experience and increase conversions.

Conclusion

The marketing funnel analysis successfully identified customer conversion behavior, major drop-off points, and

high-performing business areas. The insights generated from this project can help businesses improve conversion strategies, optimize marketing campaigns, and enhance customer experience. This project demonstrates practical business analytics, marketing analytics, and data visualization skills using Python and Power BI.